

Article

Creating specialized classes for fire risk using machine learning methods.

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Abstract

This study explores various machine learning techniques for assessing forest fire risk. The data used in this study have been selected from three different areas in Greece over the past 25 years (2000 - 2024). The first area of interest is Attica, the second area is eastern Macedonia and Thrace, and the third area is Crete. For each of the aforementioned areas, a separate data file was created, an analysis was carried out, and specialized definitions - classes were created for the forest fire risk occurrence. This study utilized a range of traditional machine learning techniques alongside novel methods based on Grammatical Evolution, including automated feature construction from existing attributes and the automatic generation of classification rules.

Keywords: Fire risk; Machine learning; neural networks; Grammatical evolution.

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